

A STUDY ON THE PERFORMANCE ENHANCEMENT OF CANOES USING HYDROFOIL

A MAJOR PROJECT REPORT

Submitted by

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(Under Section 3 of UGC Act, 1956)

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ABSTRACT

In the dynamic and demanding realm of canoeing, where every glide determines success, elevating lift while minimizing overall drag is key to improving performance. This study investigates the enhancement of canoe performance by retrofitting hydrofoils to optimize lift and reduce drag, thereby enhancing speed on water surfaces. Through literature review and project scoping, the selection and design of the hydrofoil, along with the integration of winglets, emerged as the central strategy for performance enhancement. Utilizing SolidWorks, a precise CAD model of a canoe was designed. CFD (Computational Fluid Dynamics) simulations were run on ANSYS Fluent and the lift and total drag forces were observed for the canoe before and after retrofitting with hydrofoils.

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CHAPTER 1

INTRODUCTION

Understanding the dynamics of fluid flow around and within objects is fundamental in enhancing performance, particularly in waterborne sports like canoeing. The manipulation of water flow around a canoe significantly impacts its speed, maneuverability, and efficiency. Hence, optimizing the flow around and beneath the canoe's hull becomes imperative for constructing the most efficient watercraft possible. In competitive canoeing, achieving maximum speed while minimizing resistance is paramount. This entails comprehending and optimizing the hydrodynamic behavior of the canoe, particularly with the incorporation of hydrofoils. These hydrofoils, analogous to wings in aerodynamics, aim to enhance lift and reduce drag, thereby improving overall performance on water surfaces. This study delves into the integration of hydrofoils to augment canoe performance, aiming to maximize lift and minimize drag in accordance with the regulations of competitive canoeing. By meticulously examining existing literature and project requirements, the optimal design and placement of hydrofoils will be determined. Through rigorous testing and analysis, the effectiveness of hydrofoils in enhancing canoe performance will be evaluated, providing insights into the intricacies of hydrodynamic optimization in canoeing.

Aerodynamics of a Canoe

Canoe aerodynamics play a crucial role in its performance on water. Unlike cars, canoes experience hydrodynamic forces instead of aerodynamic ones. The shape of the hull affects drag, with sleek designs reducing resistance and enhancing speed. Stability is vital to prevent capsizing, so canoe designers prioritize minimizing lift forces. Additionally, features like keels and chines aid in maintaining course stability and maneuverability. Overall, optimizing canoe hydrodynamics involves balancing drag reduction with stability and maneuvering requirements for efficient and enjoyable paddling experiences.

Drag Force:

The drag force on a canoe, caused by resistance from the water, limits its speed and efficiency. However, hydrofoils, which are wing-like structures submerged beneath the water's surface, can significantly reduce drag by lifting the canoe partially out of the water. By minimizing the contact area between the canoe and the water, hydrofoils decrease drag, allowing the canoe to glide more smoothly and efficiently through the water, ultimately enhancing speed and performance.

 C_l & C_d Coefficients:

A dimensionless coefficient known as the lift coefficient (C_l) connects the lift produced by a lifting body to the fluid velocity, surrounding fluid density, and related reference area. Aerodynamicists use the lift coefficient as a straightforward tool to model all of the intricate relationships between shape, inclination, and certain flow conditions and lift. The ratio of the lift force to the force generated by the dynamic pressure times the area is expressed by this coefficient. Lift coefficient is frequently determined through wind tunnel testing or computer-generated models (CFD). The lift coefficient has a constant value within specific operating conditions and in the absence of stall. The lift or downforce generated is then proportional to the square of airspeed and can be calculated using the following formula

$$L = \frac{1}{2} C_l \rho u^2 A$$

Where,

u = Flow speed;

ρ = Fluid density;

A = Pertinent plan area;

L = Lift force

In terms of aerodynamics, the drag coefficient (C_d) is a commonly used metric in design. An automobile's drag coefficient affects how well it moves through the surrounding air. At

higher speeds, aerodynamic drag becomes crucial because it rises with the square of speed. Performance in terms of speed is enhanced by lowering its drag coefficient. The way that the body is shaped affects C_d . Drag force is proportional to air density and the square of the relative speed between the air and the object, and it varies with C_d and flow direction, object size, speed, and air density and viscosity.

The drag equation can be used as one means of expressing this:

$$D = \frac{1}{2} C_d \rho u^2 A$$

Where,

u = Flow speed;

ρ = Fluid density;

A = Pertinent plan area;

L = Lift force

Depending on the kind of drag coefficient being measured, reference area A will change. The projected frontal area of a vehicle serves as the reference area for many objects. Depending on where the cross section is taken, this might not be the vehicle's cross-sectional area. The nominal wing area is used as the reference area for airfoils.

The body shape of a vehicle has an impact on its drag coefficient. The wake must be narrow in order for the boundary layer surrounding the body to stay as attached to the body's surface as possible and produce a low drag coefficient. A broad wake is produced by a high form drag. It will shift the boundary layer from laminar to turbulent if the flow surrounding the body has a high enough Reynolds number. Reynolds numbers increase with larger objects, higher velocities, and lower viscosities.

The coefficient of drag is also influenced by a number of other factors. For example, some race cars may have higher drag coefficients than others, but this is only to offset the lift they produce.

CHAPTER 2

LITERATURE SURVEY

Chen, Shuling (2013)^[13] explores the intricate dynamics of hydrofoil crafts, shedding light on their hydrodynamic characteristics and performance. Through detailed analysis and experimentation, the study delves into various aspects of hydrofoil behaviour, including lift generation, drag reduction, stability, and maneuverability. Key findings highlight the significant role of hydrofoil design in optimizing performance, with foil shape, angle of attack, and surface roughness playing crucial roles. The paper elucidates how hydrofoils generate lift by exploiting Bernoulli's principle and the phenomenon of dynamic pressure variation.

Daniel Savitsky, 1964^[15] provides a comprehensive examination of the hydrodynamic principles governing the behavior and performance of planing hulls. Through theoretical analysis, computational modeling, and experimental validation, the study elucidates the key factors influencing planing hull dynamics, including hull shape, weight distribution, and surface interaction. Moreover, the paper investigates the effects of various design parameters on planing hull performance, such as hull deadrise angle, chine geometry, and trim angle. It also addresses practical considerations, such as the impact of hull loading, propulsion systems, and operating conditions on planing efficiency and stability.

Bal, 2003^[5] introduces a sophisticated computational model designed to simulate the behavior of hydrofoils under cavitating conditions within a wave tank environment. By combining numerical methods with advanced fluid dynamics simulations, the study aims to accurately predict the complex phenomena associated with cavitation, which can significantly impact the performance and stability of hydrofoil systems. Key components of the numerical wave tank model include the incorporation of multiphase flow dynamics, which account for the formation and collapse of vapor bubbles around the hydrofoil surfaces. The model also considers the interaction between the cavitation-induced flow features and the free surface waves within the tank, offering a comprehensive representation of the hydrodynamic environment.

Alec, 2019^[2] investigates the dynamic stability characteristics of a hydrofoiling sailing boat through computational fluid dynamics (CFD) simulations. The study focuses on understanding how various factors influence the stability of the hydrofoil system during sailing manoeuvres, such as tacking and gybing. Using CFD, the authors model the hydrodynamic forces acting on the hydrofoils and hull of the sailing boat under different sailing conditions. They analyze the effects of parameters like foil configuration, hull shape, sail setting, and hydrofoil angle of attack on the boat's stability and performance.

Azcueta R, 2003^[3] presents a comparative analysis of computational fluid dynamics (CFD) simulations applied to planing crafts under both steady and unsteady flow conditions. The study aims to evaluate the accuracy and reliability of Reynolds-averaged Navier-Stokes equations (RANSE) simulations in predicting the hydrodynamic behavior of planing hulls. The authors first describe the numerical methodologies employed in both steady and unsteady RANSE simulations, highlighting the key assumptions and governing equations utilized to model the flow around planing crafts. They discuss the challenges associated with capturing complex flow phenomena, such as hull-water interaction, wave generation, and spray formation, which are critical for accurate simulation results.

Bal, 2001^[6] investigates the behavior of cavitating hydrofoils in both two-dimensional (2-D) and three-dimensional (3-D) configurations, considering the presence of a free surface. Through numerical analysis, the study aims to elucidate the complex interactions between the hydrofoil, cavitation bubbles, and the water surface, which are crucial for understanding and optimising hydrofoil performance. The authors describe the numerical methodology employed, which involves solving the Navier-Stokes equations coupled with a cavitation model to simulate the formation, growth, and collapse of vapor bubbles around the hydrofoil surfaces. The simulations consider both 2-D and 3-D geometries to capture the full three-dimensional flow behaviour, including spanwise variations and surface effects.

Bowker J, 2018^[11] explores the complex interactions between a flapping foil propulsion system and the dynamic response of a wave-powered vessel. By integrating fluid-structure interaction analysis with numerical simulations, the study investigates how the motion of the vessel and the propulsion system influence each other, ultimately affecting overall

performance. Key findings highlight the importance of achieving synchronisation between the flapping motion of the foils and the natural frequency of the vessel to maximise propulsion efficiency. The paper also explores the trade-offs between different design choices, such as foil size, shape, and flapping frequency, in optimising performance under varying wave conditions.

Niklas K, 2019^[28] explores the feasibility and reliability of using full-scale computational fluid dynamics (CFD) simulations as an alternative method for predicting ship resistance compared to traditional towing tank experiments. The study investigates the accuracy of CFD simulations in capturing the hydrodynamic forces acting on a ship hull and their impact on resistance, offering a rational approach to supplement or replace physical model testing. Through validation against experimental data from towing tank tests, the study evaluates the accuracy and reliability of the CFD predictions in estimating ship resistance across a range of operating conditions and hull geometries.

Niklas K, 2019^[29] investigates the hydrodynamic performance of a ship with two different hull forms – a V-shaped bulbous bow and an X-bow – through full-scale computational fluid dynamics (CFD) seakeeping simulations. The study aims to evaluate the effectiveness of the X-bow hull design in improving seakeeping behavior compared to the traditional V-shaped bulbous bow. Key findings highlight the advantages of the X-bow hull form in reducing wave impact and improving seakeeping characteristics compared to the V-shaped bulbous bow. The paper discusses how the distinctive shape of the X-bow minimizes slamming and pitch motions, leading to smoother sailing and enhanced crew comfort in rough seas.

Niklas K, 2023^[30] examines the implementation and performance of retractable bow hydrofoils as a retrofit solution for improving ship efficiency and stability. The case study investigates the practicality and effectiveness of this retrofitting approach through a detailed analysis of a specific vessel. The paper describes the methodology used in retrofitting the ship with retractable bow hydrofoils, including the design process, installation procedure, and integration with existing ship systems. It discusses the rationale behind choosing this retrofitting solution and the expected benefits in terms of fuel efficiency, manoeuvrability, and seakeeping performance. Key findings from the case

study highlight the positive impact of retractable bow hydrofoils on ship performance. The hydrofoils are shown to reduce resistance, decrease fuel consumption, and enhance stability, particularly in adverse weather conditions. The paper discusses how the retrofitting process was successfully implemented and the challenges encountered during installation and testing.

Putranto, 2017^[31] investigates the influence of hydrofoil geometry and angle of attack on lift-to-drag ratio and form factor. The study employs analytical methods to analyze the hydrodynamic performance of hydrofoils under various conditions. The paper outlines the methodology used to assess the lift-to-drag ratio and form factor of hydrofoils, considering factors such as foil shape and angle of attack. Through theoretical analysis and mathematical modeling, the study explores how changes in these parameters affect the hydrofoil's aerodynamic characteristics.

Qu, 2020^[32] explores the application of computational fluid dynamics (CFD) to simulate the hydrodynamic performance of a yacht's hydrofoil. The study aims to analyse the flow characteristics around the hydrofoil and assess its lift, drag, and overall performance. The methodology used for CFD simulation includes grid generation, boundary conditions, and turbulence modeling. By solving the Navier-Stokes equations, the study models the flow behaviour around the hydrofoil under various operating conditions and angles of attack.

H. Liang, 2013^[18] presents an analytical model for analysing the hydrodynamic behaviour of a three-dimensional hydrofoil equipped with winglets, operating beneath a free surface. The study aims to provide a comprehensive understanding of the flow interactions between the hydrofoil, winglets, and free surface, considering both lift generation and wave-making effects. The paper outlines the theoretical framework used to develop the analytical model, which involves solving the potential flow equations in three dimensions, accounting for the presence of the free surface and the geometric features of the hydrofoil and winglets. The study considers various factors, such as hydrofoil geometry, winglet configuration, and operating conditions, to assess their influence on hydrodynamic performance.

Brown D. K, 1984^[12] examines the performance and capabilities of the hydrofoil vessel HMS Speedy. The study aims to provide a comprehensive assessment of the hydrofoil's

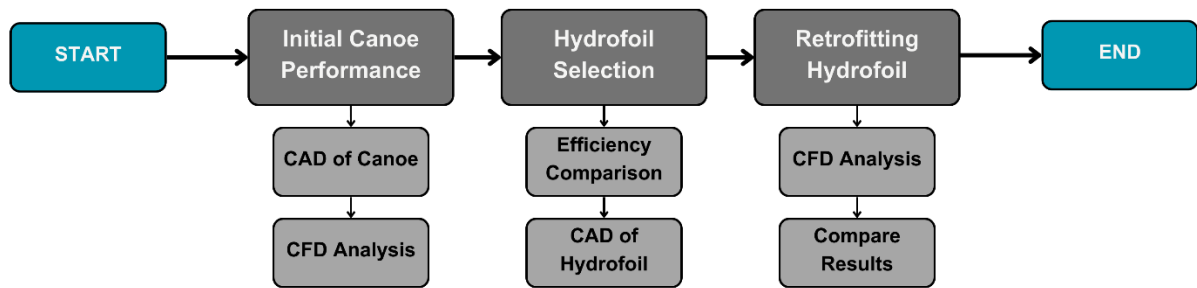
design, operational characteristics, and effectiveness in fulfilling its intended roles. Key findings from the evaluation highlight the hydrofoil's strengths and limitations in terms of speed, agility, and efficiency. The paper discusses how HMS Speedy's hydrodynamic design enables it to achieve high speeds and manoeuvrability, making it well-suited for certain maritime operations.

A. Li, 2022^[41] explores the application of computational fluid dynamics (CFD) in optimising a hydrofoiling concept for sea transport. The study aims to enhance the efficiency, stability, and performance of hydrofoil-based vessels through advanced numerical simulations. The paper outlines the methodology used for CFD optimization, which involves modeling the hydrofoil geometry, flow dynamics, and operational conditions using state-of-the-art computational tools. By iteratively refining the hydrofoil design and analysing its hydrodynamic characteristics, the study seeks to identify the optimal configuration for maximising performance. Key findings from the CFD optimization highlight the importance of factors such as foil shape, angle of attack, and hydrodynamic interactions in determining hydrofoil performance. The study discusses how changes in these parameters can influence lift generation, drag reduction, and the overall efficiency of the hydrofoil system.

CHAPTER 3

OBJECTIVES AND METHODOLOGY

Given the impact of viscous drag on a canoe, a hydrofoil will help reduce the drag and improve the canoe's performance. With the help of a literature survey, various hydrofoils were chosen and tested to see which one was best suited for the canoe. This process was done by initially determining the effect of drag on the canoe, designing a hydrofoil, and retrofitting the hydrofoil to the canoe to determine the improvement. The design speed of the project is set to the average speed of a canoe with a world record time of 37.446 seconds in a 200 m sprint; this equates to an average speed of 5.33 m/s or 19.2 km/hr.



Flowchart of Project

3.1 3D Modelling

When preparing a CFD model, it is required to import a geometry from a computer-aided design (CAD) software for simulation analysis. This project utilised SolidWorks (CAD Software) to design the canoe and the hydrofoil before they were imported to the CFD software.

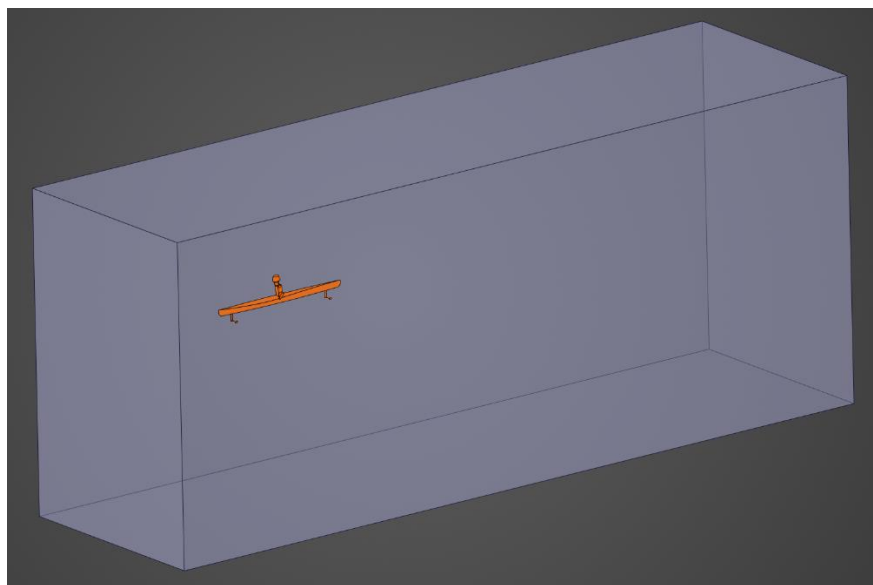
The modelling process for the canoe involves using reference images of canoes used at canoe sprint events. From the reference images, guide lines are created, and the dimensions of the K1 canoe are inputted (5.2m length, 0.4m width). For the athlete, we used the CAD model of a 95th Percentile Male obtained from GrabCAD. The hydrofoil is designed by inputting a 2D foil and extruding the sketch. Strut of the hydrofoil is also designed by inputting a 2D foil; the 2D foil is imported with a .txt file that contains the coordinates.

3.2 CFD Simulation

Computational fluid dynamics (CFD) analyses the flow of fluid using the Navier-Stokes equation. It is applied in engineering fields to solve problems such as aerodynamics and hydrodynamics. ANSYS Fluent is used to run the CFD simulations. The Fluid Domain, Mesh and Boundary Conditions are defined in ANSYS Fluent. The simulation is using a multiphase model of phases air and water. The only difference between the initial canoe model and retrofitted canoe model is the addition of the hydrofoil to the retrofitted canoe. Hence, the models are similar, the settings discussed below are same for both models.

3.2.1 Creation of Fluid Domain

The fluid domain helps define the computational domain, which is filled by one or more fluids. To ensure boundaries of the fluid domain do not affect with the simulation, the boundaries are kept far from the geometry. The distance from the upstream inlet boundary of the domain is 1.5 times the characteristic length in front of the geometry. Respectively, the distance from the downstream outlet boundary is 3 times the characteristic length behind the geometry. The distance from the top and bottom boundary walls is 1.5 times the characteristic length of the canoe above and below respectively. A plane along the symmetry of the canoe is created to divide the fluid domain into two halves, this helps by reducing computing time.



Fluid Domain

3.2.2 Mesh Creation

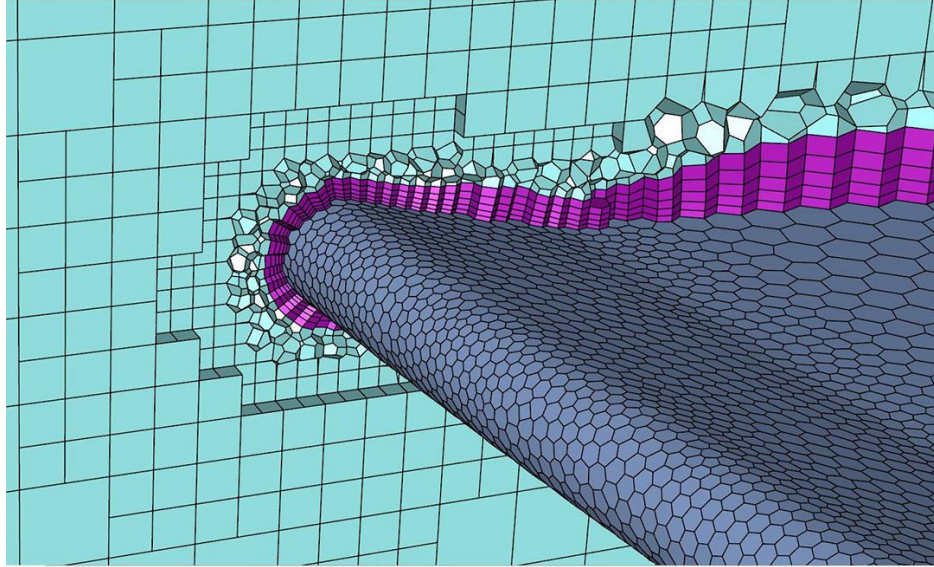
When a canoe is in water, there are multiple sources of drag. From the wake caused by the athlete to the viscous drag from the canoe caused by the high density of water. It has a significant impact on the canoe's performance. To predict accurately the drag, the fluid domain is discretised into small parts called elements joined together by nodes, creating a mesh. With the help of ANSYS Fluent Meshing, the mesh is further refined by adding local refinement regions around the canoe and boundary layers near the surfaces. These further refinements help by capturing critical areas with an appropriately refined mesh. To determine the boundary layer thickness, a calculator like Fluid Mechanics 101's Y+ Calculator is used.

Velocity (m/s)	Length Scale (m)
6.25	0.350
Viscosity (Pa s)	Density (kg/m ³)
0.00089	998
Target y^+ (-)	Layers (-)
1	20
Calculate	
First Layer (m)	Final Layer (m)
7.12e-6	1.77e-3
δ_{99} (m)	Max. Growth Ratio (-)
7.01e-3	1.34

Fluid Mechanics 101 Y+ Calculator

The generation of the volume mesh is done by filling the domain with Poly-Hexcore volume elements. In ANSYS Fluent Meshing, "Poly-Hexcore Volume Elements" refers to a meshing method that generates polyhedral and hexahedral elements in the volume of a computational domain. Using poly-hexcore volume elements in ANSYS Fluent Meshing allows you to strike a balance between mesh quality and computational efficiency, making it suitable for a wide range of simulation applications, especially in cases where you have

complex or irregular geometries that may be challenging to mesh with pure hexahedral elements. This method can improve the accuracy and convergence of simulations in ANSYS Fluent.

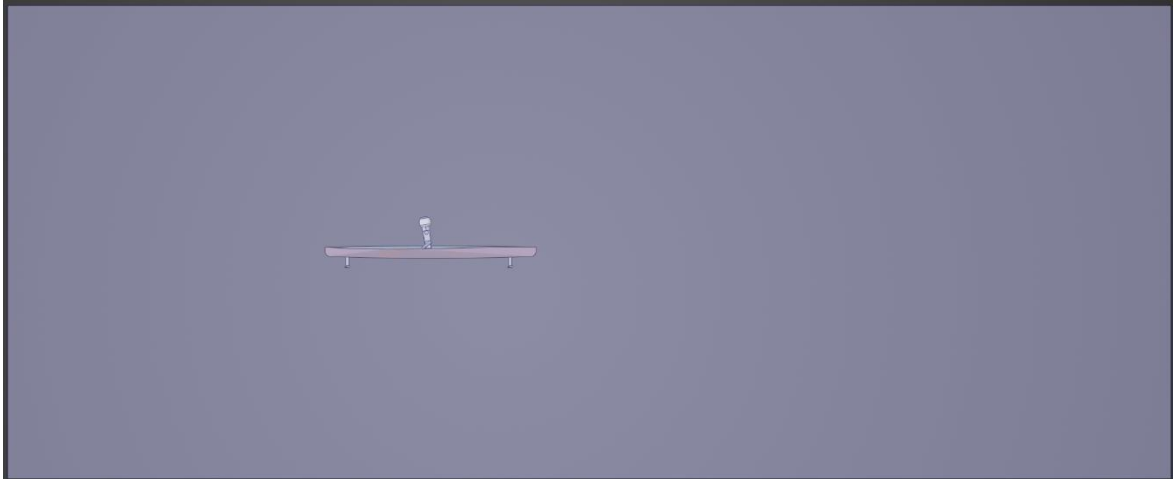


Poly-Hexcore Volume Elements

3.2.3 pre-Processing of the CFD Models

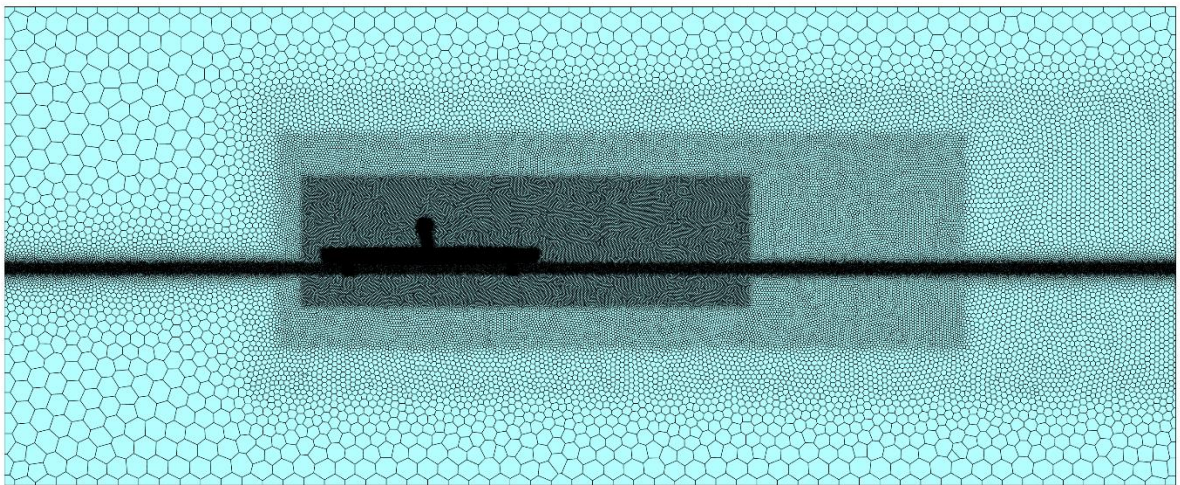
3.2.3.1 Initial Canoe Performance Simulation

The model to be tested is designed in SolidWorks and imported to ANSYS Fluent to start the pre-Processing process. For the simulation of the CFD model, the Fluid Domain is created; it extends 15600 mm to the rear of the canoe, 7800 mm to the front, 5200 mm to the sides, and 7800 mm to the top and bottom of the canoe.



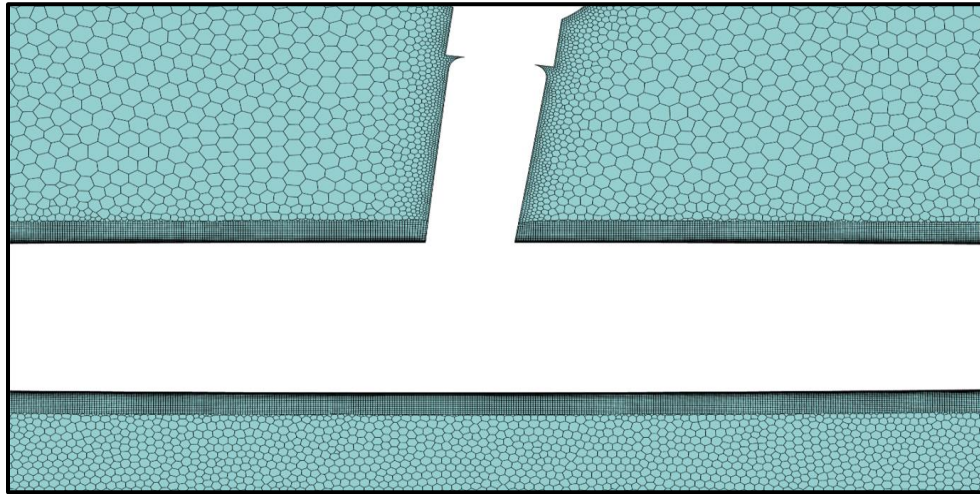
Initial Canoe - Fluid Domain

For the surface mesh of the model, ANSYS Fluent Meshing is used to create elements with a minimum surface mesh size of 2 mm and a maximum surface mesh of 512 mm. Four local refinement regions are defined to refine the mesh; three of the refinement regions are around the canoe, with maximum element sizes of 32 mm, 64 mm, and 128 mm, respectively. To accurately predict the multiphase, an additional refinement region is defined at the water level to get an accurate prediction of the multiphase.



Initial Canoe - Surface Mesh

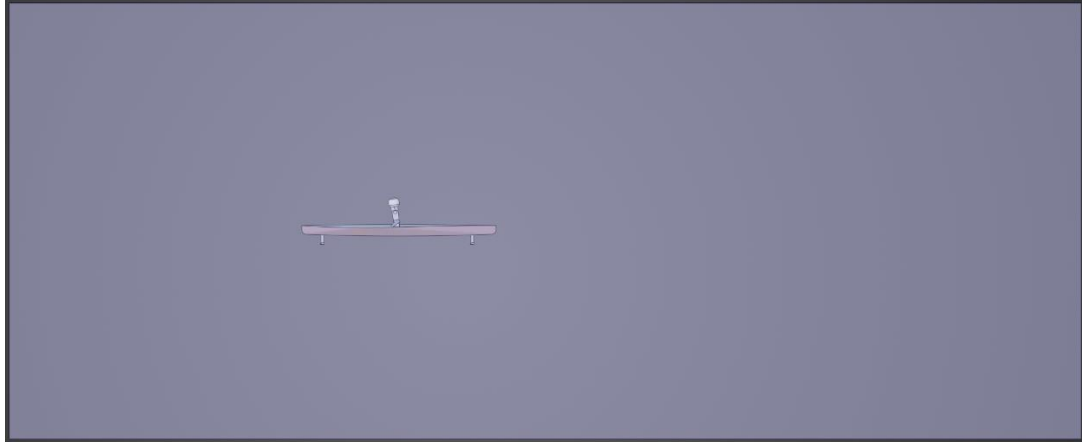
After the creation of the surface mesh, the boundary layer is created. Y^+ (Boundary Layer Height) for a characteristic length of 5.2 m and a velocity of 5.33 m/s is about 1.24×10^{-2} m. So, a 10-layered boundary is created with a growth ratio of 1.49. Finally, the volume mesh is created with poly-hexcore elements having a minimum size of 2 mm and a maximum size of 512 mm.



Initial Canoe - Surface Mesh Boundary Region Close-Up

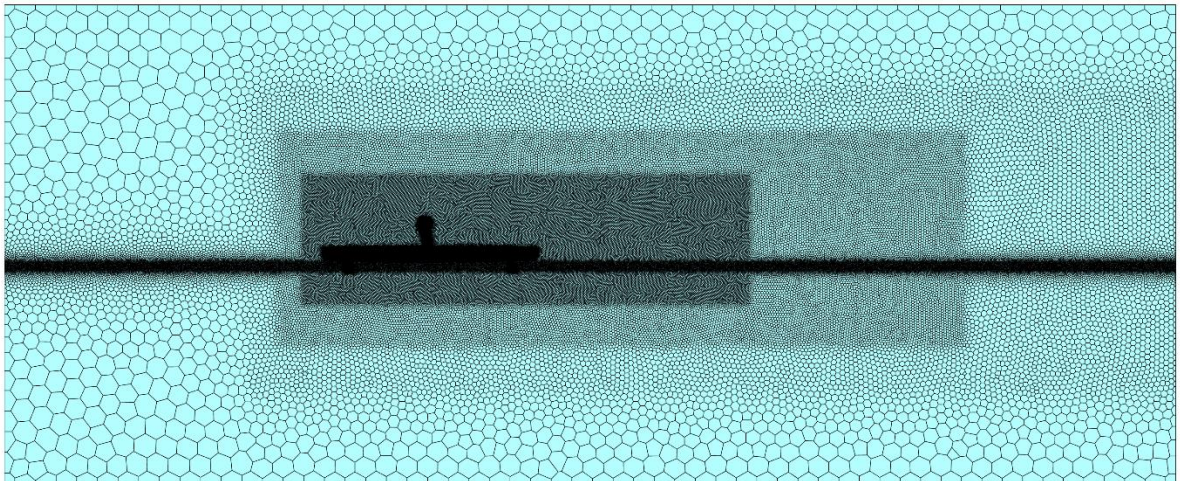
3.2.3.2 Retrofitted Canoe Performance Simulation

Similar to the initial canoe, but the hydrofoil is added to the canoe. The model to be tested is designed in SolidWorks and imported to ANSYS Fluent to start the pre-Processing process. For the simulation of the CFD model, the Fluid Domain is created; it extends 15600 mm to the rear of the canoe, 7800 mm to the front, 5200 mm to the sides, and 7800 mm to the top and bottom of the canoe.



Retrofitted Canoe - Fluid Domain

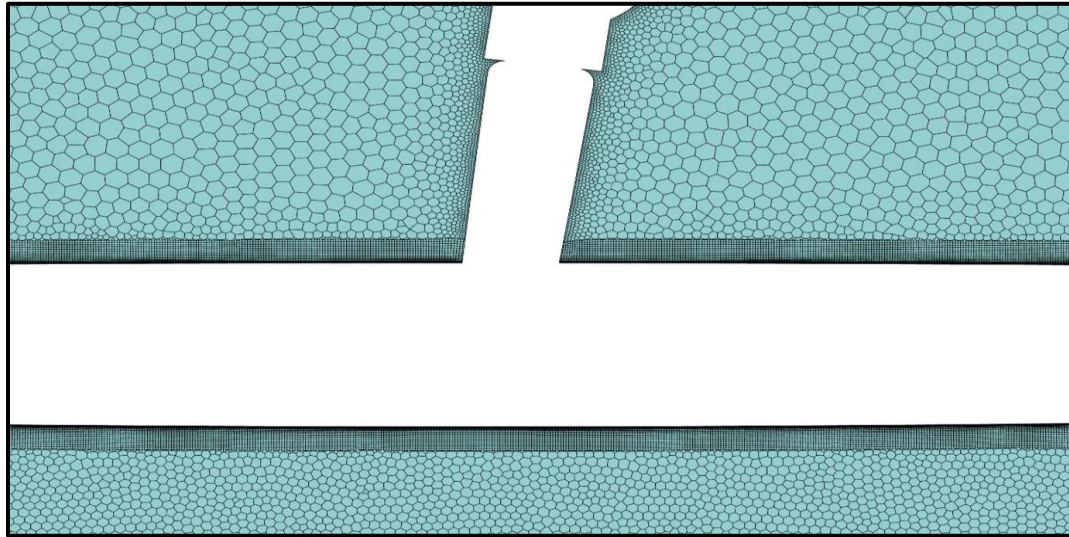
For the surface mesh of the model, ANSYS Fluent Meshing is used to create elements with a minimum surface mesh size of 2 mm and a maximum surface mesh of 512 mm. Four local refinement regions are defined to refine the mesh; three of the refinement regions are around the canoe, with maximum element sizes of 32 mm, 64 mm, and 128 mm, respectively. To accurately predict the multiphase, an additional refinement region is defined at the water level to get an accurate prediction of the multiphase.



Initial Canoe - Surface Mesh

After the creation of the surface mesh, the boundary layer is created. Y^+ (Boundary Layer Height) for a characteristic length of 5.2 m and a velocity of 5.33 m/s is about 1.24×10^{-2} .

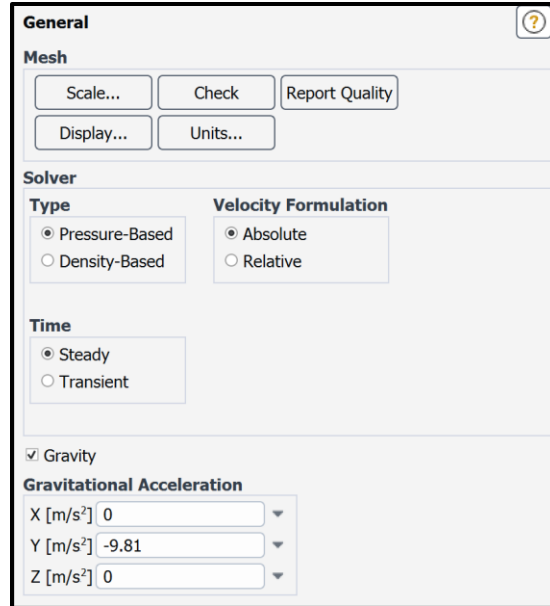
m. So, a 10 layered boundary is created with a growth ratio of 1.49. Finally, the volume mesh is created with poly-hexcore elements having a minimum size of 2 mm and maximum size of 512 mm.



Initial Canoe - Surface Mesh Boundary Region Close-Up

3.2.4 Solving

After the pre-Processing is completed and all the models are successfully meshed, the CFD cases are ready to be solved, using ANSYS Fluent. In this section, numerical methods and parameters are defined. Parameters include boundary conditions, multiphase models, turbulence models, convergence criteria, and the number of iterations. For the simulation of the canoe in water, it is assumed that the phases are incompressible and isothermal.



Ansys Fluent - General Settings

3.2.4.1 Boundary condition

As the simulation environment is a multiphase model, its inlet is a pressure inlet, and its outlet is a pressure outlet. The fluid's direction is determined by the inlet and outlet. To avoid added complexity, top, bottom, and side walls are defined as walls with zero shear.

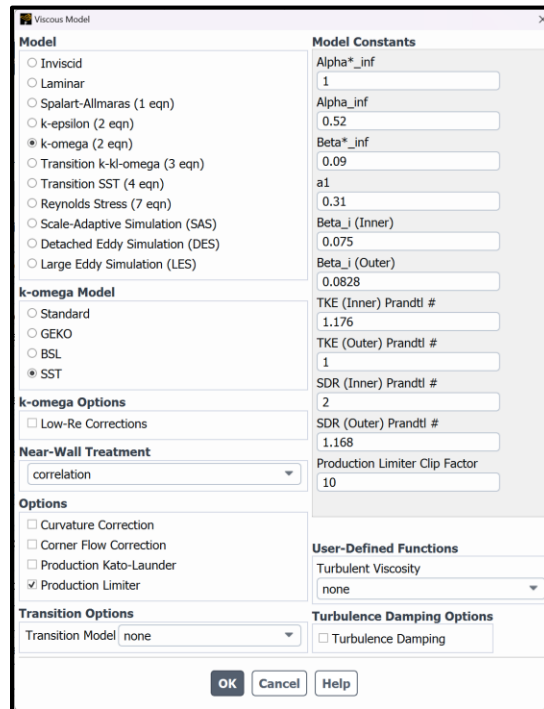
Boundary Names	Type	Value
Inlet	Pressure Inlet (Multiphase)	5.33 m/s
Outlet	Pressure Outlet (Multiphase)	0 Pa
Side	Wall	No slip wall
Top	Wall	No slip wall
Bottom	Wall	No slip wall
Symmetry	Symmetry Plane	No slip wall

Boundary Conditions of CFD Simulation

3.2.4.2 Turbulence Model

The SST $k-\omega$ turbulence model is used for solving the mesh. The most common model used to capture the effect of turbulent flow condition, has great accuracy in the boundary

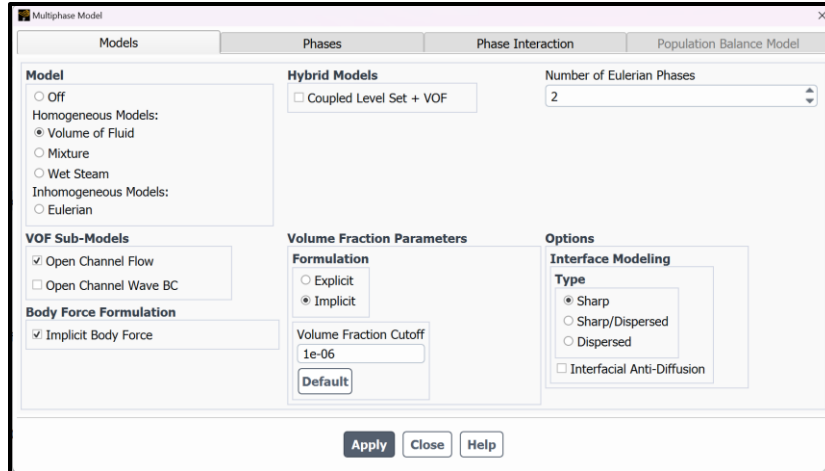
layer region close to the wall. This model can simulate the effect of viscous drag due to water on canoes.



Ansys Fluent – Turbulence Model

3.2.4.2 Multiphase Model

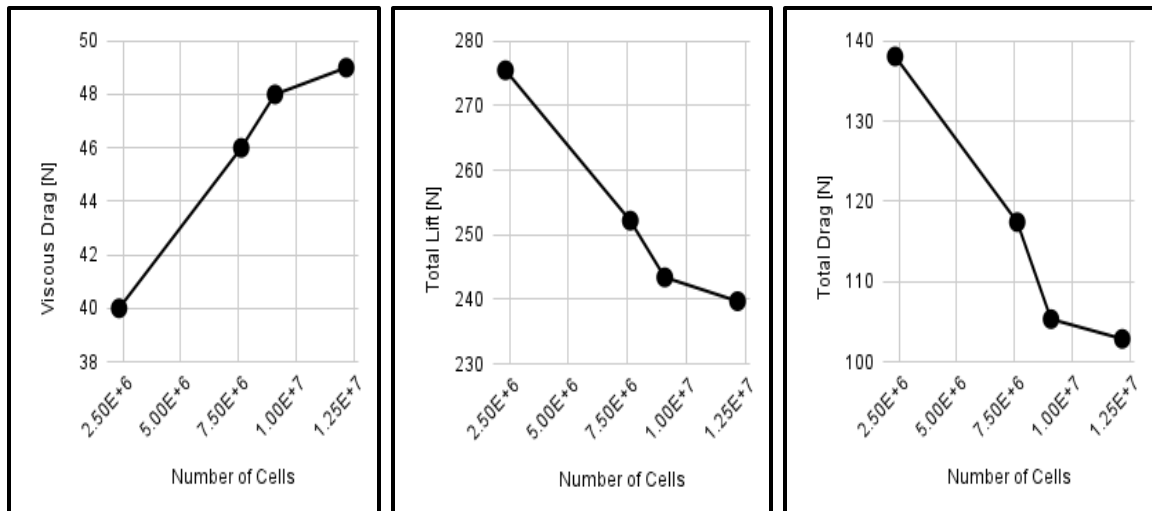
A multiphase model helps in simulating the real-world conditions affecting the canoe. A VOF model is used to simulate the conditions. In the VOF model, two or more immiscible fluids interacting with each other are simulated. It can accurately track the interference of the phases. VOF is extensively used in various industrial sectors.



Ansys Fluent – Multiphase Model

3.2.5 Mesh Independence Study

To verify that the mesh is independent of the results of the simulation. The mesh model is tested for decreasing element size, causing the number of cells in the mesh to increase. From the graphs, we get to see that a mesh with $1.00e^7$ cells has a change of less than 1%, confirming that the results are mesh-independent.



(a)

(b)

(c)

Mesh Independence Study

3.3 Hydrofoil Selection

We can choose the most efficient airfoil by considering the airfoils at several angles of attack and a specific Reynolds Number. The project has a design speed of 5.33 m/s or 19.2 km/hr, density of water (ρ) is 998.2 kg/m^3 , dynamic viscosity (μ) is 0.00089 Pa.s , and the characteristic length (l) is 0.1 m, which is equivalent to the chord length of the hydrofoil. The Reynolds number is calculated approximately to 1,000,000. To calculate the efficiency of the airfoils, polar data is obtained from Xfoil for Reynolds number $1.00e^6$.

$$Re = \frac{\rho u L}{\mu}$$

Where,

Re = Reynolds number;

ρ = Fluid Density;

u = Flow Speed;

L = Characteristic Length;

μ = Dynamic Viscosity of the Fluid

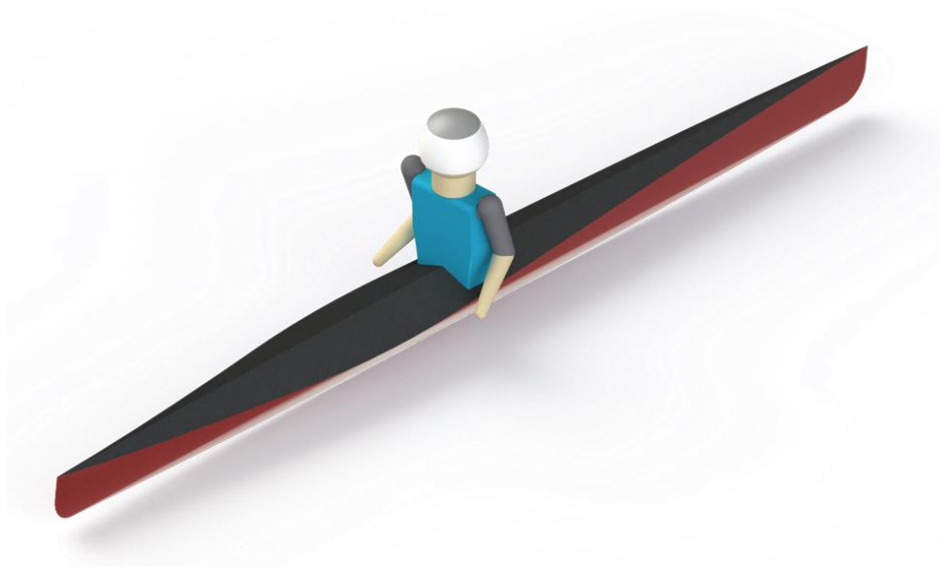
From the literature survey, 8 airfoils are selected, NACA 4412, NACA 63(2)-615, NACA 64-015, Eppler 908, Eppler E836, Eppler 837, Eppler 838 and YS-915. The highest efficient airfoil is selected, for angle of attack (AOA) varying from 0° to 8° .

CHAPTER 4

ANALYSIS

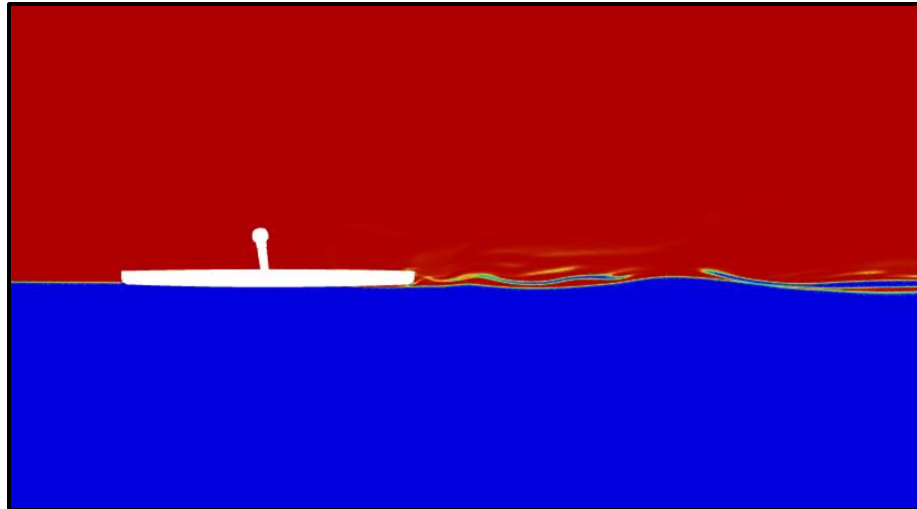
4.1 Initial Performance of Canoe

The initial part of the project is to test the canoe in water to determine the drag affecting its performance. The aim of this simulation is to determine the efficiency of the canoe without adding the retrofitted hydrofoil. It gives the baseline of the project, and the reason for the low performance of canoes is water.



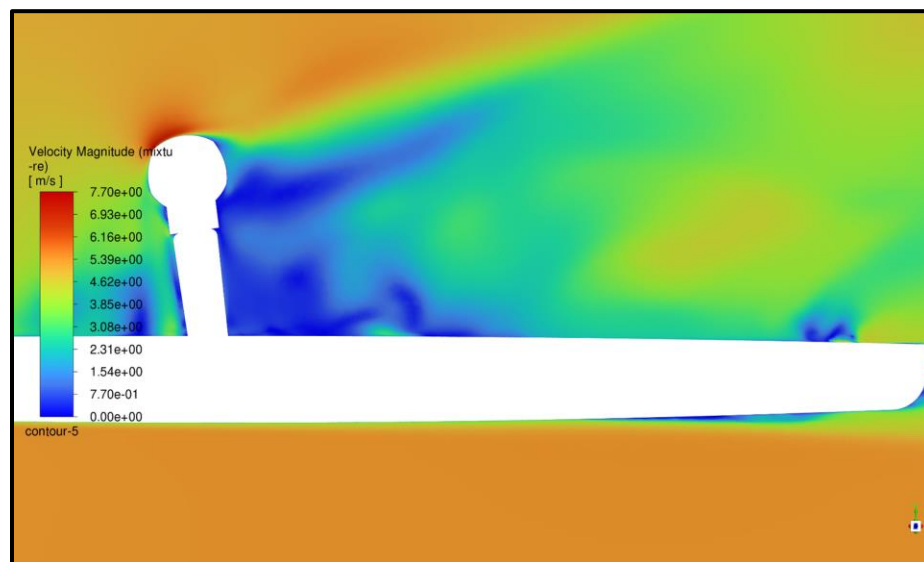
CAD of Initial Canoe

The canoe is simulated in a multiphase model at 5.33 m/s, with the water surface level intersecting with the canoe. The total drag produced by the canoe is 116.95 N, and the total lift is 236.50 N. The drag of the canoe is due to many components; the most prominent source of drag is the viscous drag of 49.81 N produced by the canoe in water. Water's high density causes high resistance for the canoe to flow. Other contributions to the drag are due to the wake caused by the athlete, but this is out of the scope of this project. The canoe experiences lift due to the athlete and the tendency of the canoe's nose to lift due to the pressure difference experienced by the canoe.

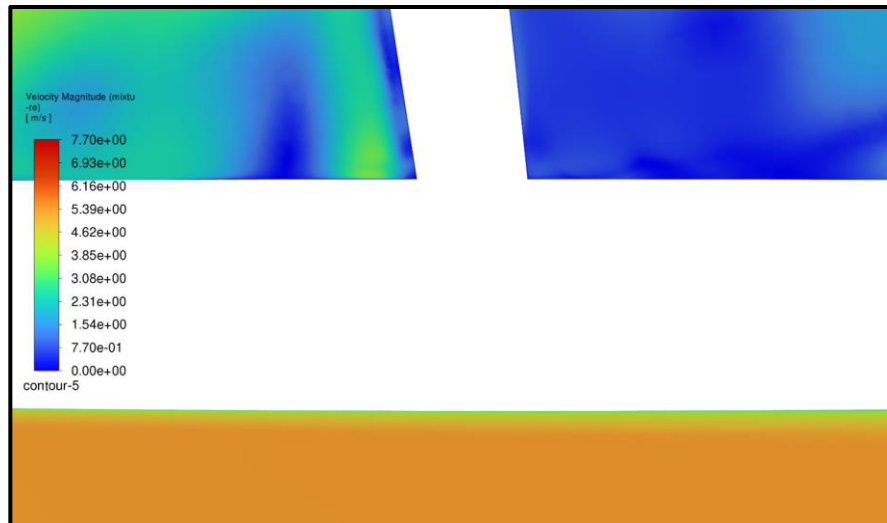


Initial Canoe – Phase Contour (Air: Red, Water: Blue)

From the velocity profile, we can see the viscous drag being produced in the boundary region of the canoe. This drag can be reduced by introducing a hydrofoil to the canoe and lifting it above the water level, reducing the viscous drag.



Initial Canoe – Velocity Contour on Symmetry



Initial Canoe – Velocity Contour [Boundary Layer Close Up]

Total Drag [N]	116.95
Viscous Drag [N]	49.81
Total Lift [N]	236.50

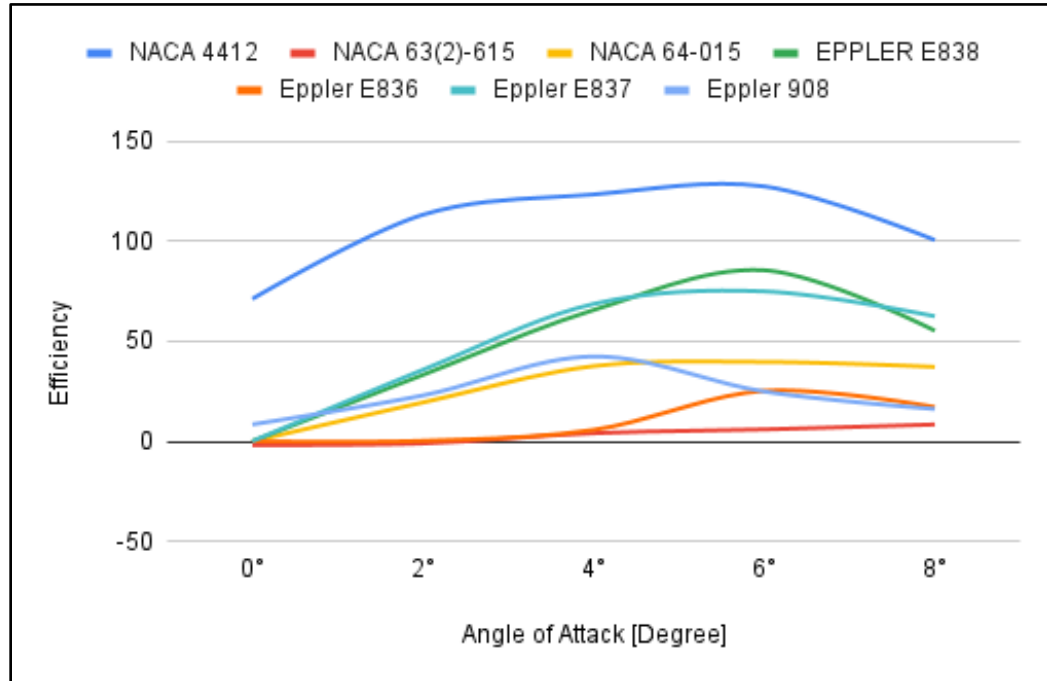
Initial Canoe Performance Results

4.2 Hydrofoil Selection

In order to properly design the hydrofoil of the canoe, it is necessary to first select the appropriate NACA airfoil to design the hydrofoil, estimate the most efficient angle of attack, and determine the dimensions and configuration of the hydrofoil(s). From the literature survey, tandem hydrofoils were found to be the optimal configuration for the hydrofoils. The requirement of the hydrofoil is to generate a lift force of 843.66 N due to the weight of the canoe, and the athlete is 86kg. For two tandem hydrofoils, each hydrofoil needs to generate around 421.83 N.

4.2.1 Comparing Efficiency v/s AOA

The first step before proceeding to the design of a complete hydrofoil is the selection of the most efficient type of NACA airfoil. Airfoils compared are NACA 4412, NACA 63(2)-615, NACA 64-015, Eppler 908, Eppler E836, Eppler E837, Eppler E838 and YS-915.



Efficiency (C_l/C_d) of Airfoils v/s Angle of Attack (AOA)

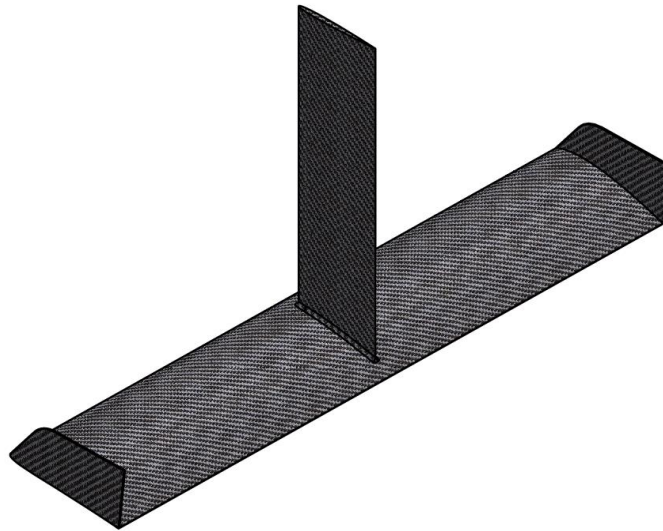
For Reynold's Number $1.00e^6$, NACA 4412 at 6° angle of attack produced the highest efficiency (C_l/C_d).

4.3 Retrofitted Canoe with Hydrofoil

With the hydrofoil selected, the final part of the project is to design the hydrofoil and test the retrofitted canoe. The tasks for the final part are to determine the dimensions of the hydrofoil, design the winglets, design the strut, and finally determine the drag produced by the retrofitted canoes. The retrofitted canoe is simulated in a multiphase model at 5.33 m/s, with the water level below the canoe.

4.3.1 Hydrofoil Optimisation

To finalise the span of the hydrofoil, three hydrofoils of different spans and the same chord length are designed and tested. Hydrofoil #1 has a chord length of 0.1 m and a span of 0.4 m; Hydrofoil #2 has a chord length of 0.1 m and a span of 0.5 m; and Hydrofoil #3 has a chord length of 0.1 m and a span of 0.6 m. The aim of this simulation is to design a hydrofoil that generates a lift force of 421.83 N.



CAD of Hydrofoil

From the three hydrofoils, Hydrofoil #2 generated the optimal amount of lift for the target lift force required. Due to a lack of time, further iteration of the hydrofoil design has not been completed. Further iterations would allow for a more accurate hydrofoil dimension for the target lift force.

Hydrofoil Span	0.4 m	0.5 m	0.6 m
Lift @Hydrofoils [N]	330.34	404.13	550.02
Drag @Hydrofoils [N]	33.62	39.64	47.66

Hydrofoil Performance v/s Span of Hydrofoil

4.3.2 Winglet Design

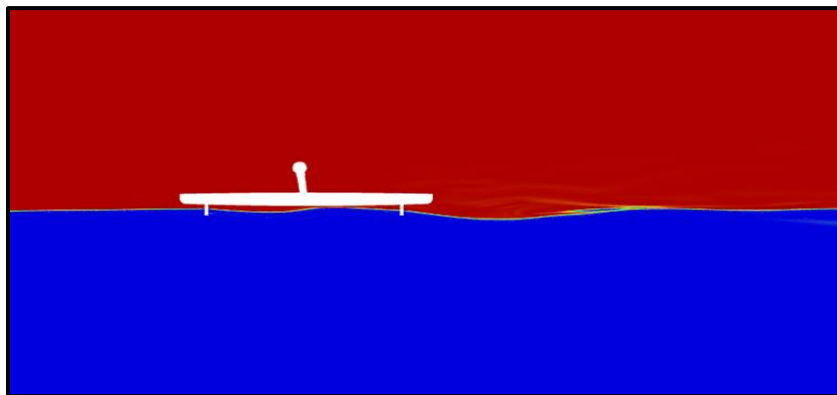
A winglet is a small vertical extension at the tips of wings or hydrofoils that serves the purpose of improving aerodynamic efficiency and reducing drag-induced energy losses. By reducing drag, winglets allow hydrofoils to move through the water with less resistance, thereby improving overall efficiency and potentially increasing speed. However, it is worth noting that the design and effectiveness of winglets on hydrofoils depend on various factors, such as hydrofoil shape, operating conditions, and intended use.

With the help of a literature survey, H. Liang compared three types of winglet configurations. For our requirements, an upper winglet on the hydrofoil is optimal.

4.3.3 Strut Design

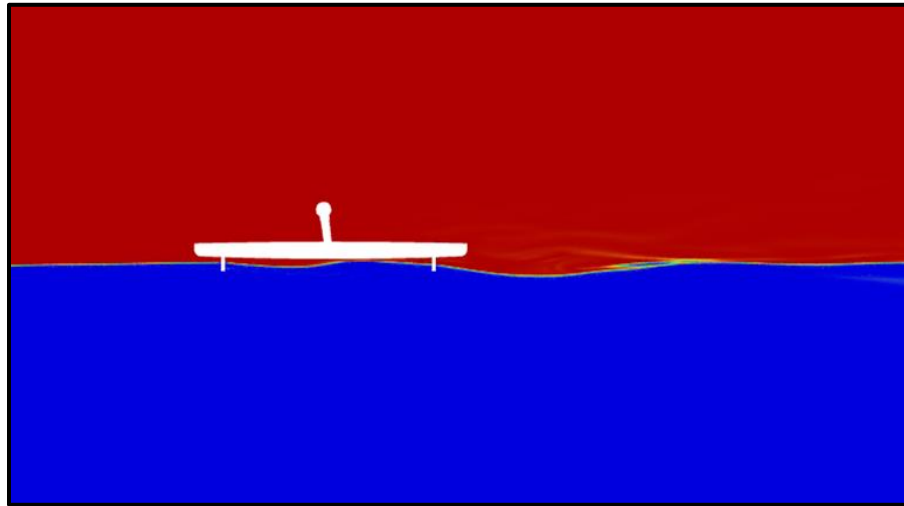
The design of hydrofoil struts is the component that connects the foil to the hull of the canoe. The strut height should be adequate to avoid contact between the canoe and the water surface during normal operation. Also, the strut should be able to keep the hydrofoil always under the water level.

Two strut types of the hydrofoil are designed and tested. The first type has a strut height of 100 mm. This initial design's purpose is to get an understanding of the spacing required from the canoe's hull, hydrofoil, and water level. However, the design leads to interaction between the water level and the hull, causing viscous drag. Also, due to waves formed by the front hydrofoil, the rear hydrofoil results in being above the water level.



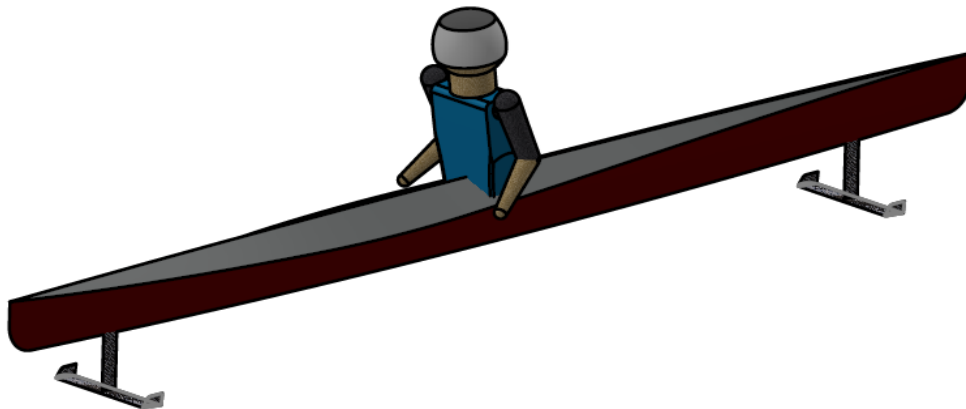
Waves formed by hydrofoil with strut length of 100 mm

The next and final design of the hydrofoil strut has a strut height of 200 m. This design has no interaction between the water level and the hull, resulting in very low viscous drag. The rear hydrofoil is also not affected by the waves caused by the front hydrofoil.



Waves formed by hydrofoil with strut length of 200 mm

4.3.4 Retrofitted Canoe Results



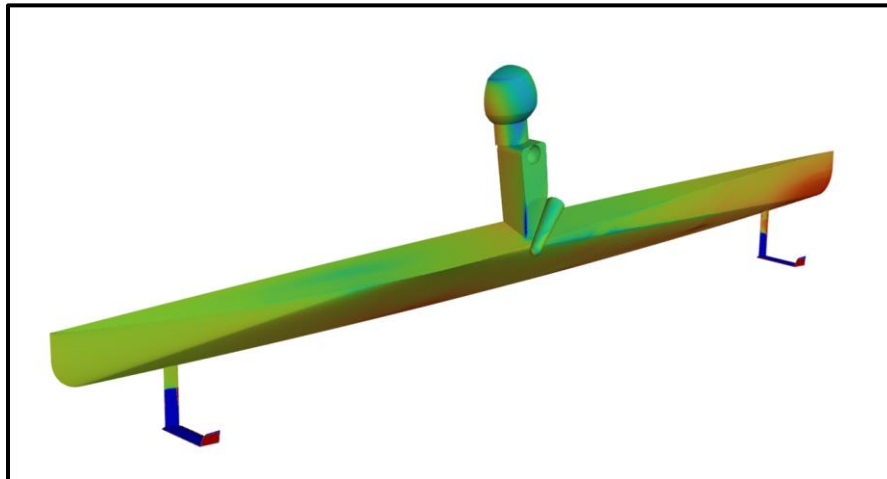
CAD of Retrofitted Canoe

The total drag produced by the retrofitted canoe is 42.85 N, and the total lift is 414.59 N. The drag produced by viscous drag is only 0.87 N due to no interaction of the canoe with the water surface. Only other contributions to the drag are due to the drag of the hydrofoil

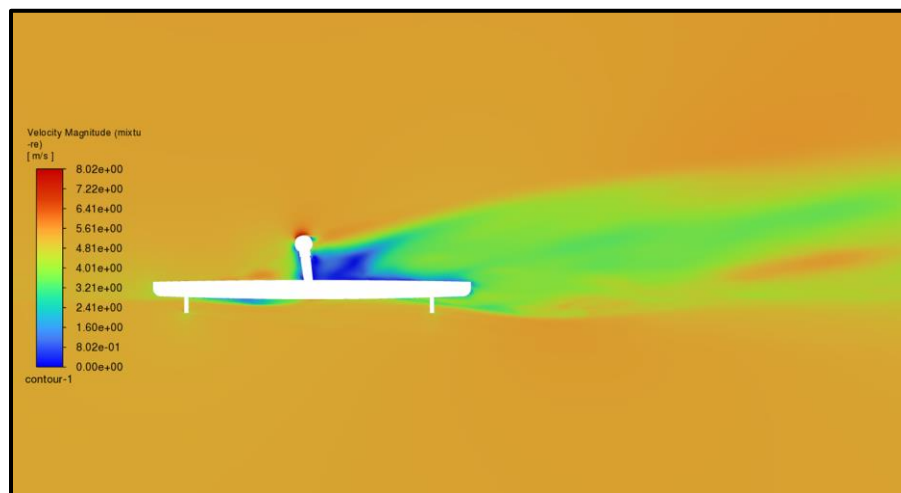
and wake caused by the athlete. The canoe experiences lift due to the hydrofoil generating an upward force of 404.13 N; the rest is due to the canoe and the athlete. The drag produced by the hydrofoils is 39.64 N due to pressure and viscous drag on them.

Total Drag [N]	42.85
Viscous Drag [N]	0.87
Total Lift [N]	414.59

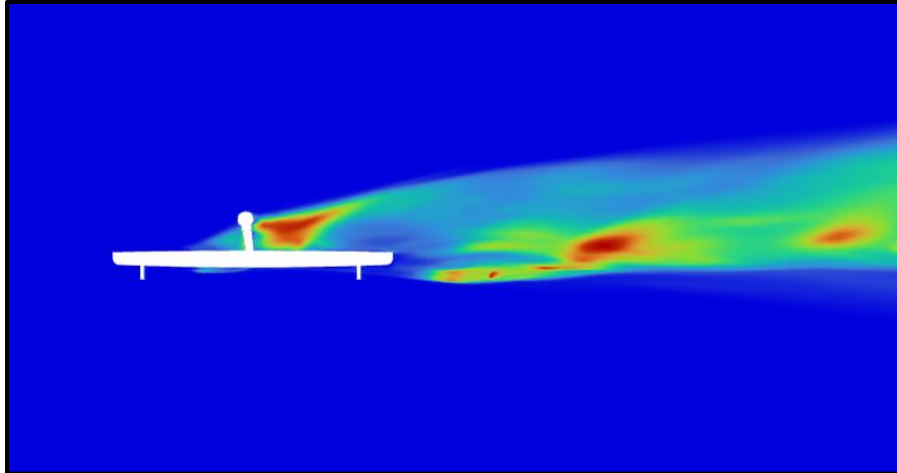
Retrofitted Canoe Performance Results



Retrofitted Canoe – Pressure Contour



Retrofitted Canoe – Velocity Contour on Symmetry



Retrofitted Canoe – Turbulent Kinetic Energy Contour

CHAPTER 5

RESULTS AND DISCUSSION

Comparing the results that occurred from the CFD simulations of the initial canoe and the retrofitted canoe, a significant drag reduction is observed. Initially, the canoe produces a drag force of 116.95 N, where 49.81 N is viscous drag on the canoe due to the interaction of water with the canoe's hull. In the retrofitted canoe, a drag force of 42.85 N is produced, and only 0.87 N is viscous drag. This reduction in the drag produced by the canoe identifies how a hydrofoil can reduce the drag produced. Another noticeable change is the increase in lift force on the canoe due to the hydrofoils.

	Initial	Final	Change
Total Drag [N]	116.95	42.85	(-) 63.3%
Viscous Drag [N]	49.81	0.87	(-) 98.2%
Total Lift [N]	236.50	414.59	(+) 75.3%

Initial v/s Retrofitted Canoe Performance

The drag produced by the hydrofoils is 39.64 N; this is due to the skin friction drag and pressure drag on the hydrofoil. A hydrofoil with a higher aspect ratio will help reduce skin friction drag.

CHAPTER 6

CONCLUSIONS

Retrofitting a hydrofoil to enhance the performance of a canoe represents a significant endeavor that merges innovation, engineering, and a passion for watercraft. Through meticulous planning, rigorous testing, and creative problem-solving, our project has achieved remarkable results. The integration of a hydrofoil system has not only increased the speed and stability of the canoe but has also opened new avenues for exploration and enjoyment on the water.

To summarise, installing a hydrofoil to improve canoe performance involves not just pushing the limits but also rethinking them. It involves challenging the boundaries of what can be achieved and embracing the principles of creativity. As we anticipate the future, we foresee significant progress in watercraft technology, fueled by our unwavering commitment to quality and our longing to fully embrace the exhilaration of navigating the wide ocean.

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